Improving the Usability of Public Transport in Oldenburg

Project Diary

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1 Introduction

1.1 Background and Motivation

Public transport in Oldenburg is used by established residents and newcomers alike. The system works, but accessibility and usability depend on familiarity with the city and its infrastructure. Newcomers' experiences reveal initial barriers that long-term residents may overlook. This project focuses on newcomers: their experience with public transport in Oldenburg, and digital measures that improve it.

1.2 Problem Statement

This project supports newcomers in Oldenburg by improving their public transport experience. "Newcomer" means anyone new to Oldenburg – moved from another region, city, or country, or just visiting – who has lived here less than two years. Newcomers may face challenges in the following areas:

- Finding the right connection
- Navigating/Routing
- Buying and choosing tickets
- Understanding fares and validity
- Payments
- Knowing which app to use
- Language
- Real-time information

This is supported by literature ([1],[2],[3]) and personal experience.

1.3 Objectives

This project follows the Interaction Design Cycle shown in Figure 1.

For each step, the objectives are:

1. **Gather Insights:** Identify what issues people face in public transport and which apply to newcomers.
2. **Compile Requirements:** Distill findings into concrete requirements for the system.

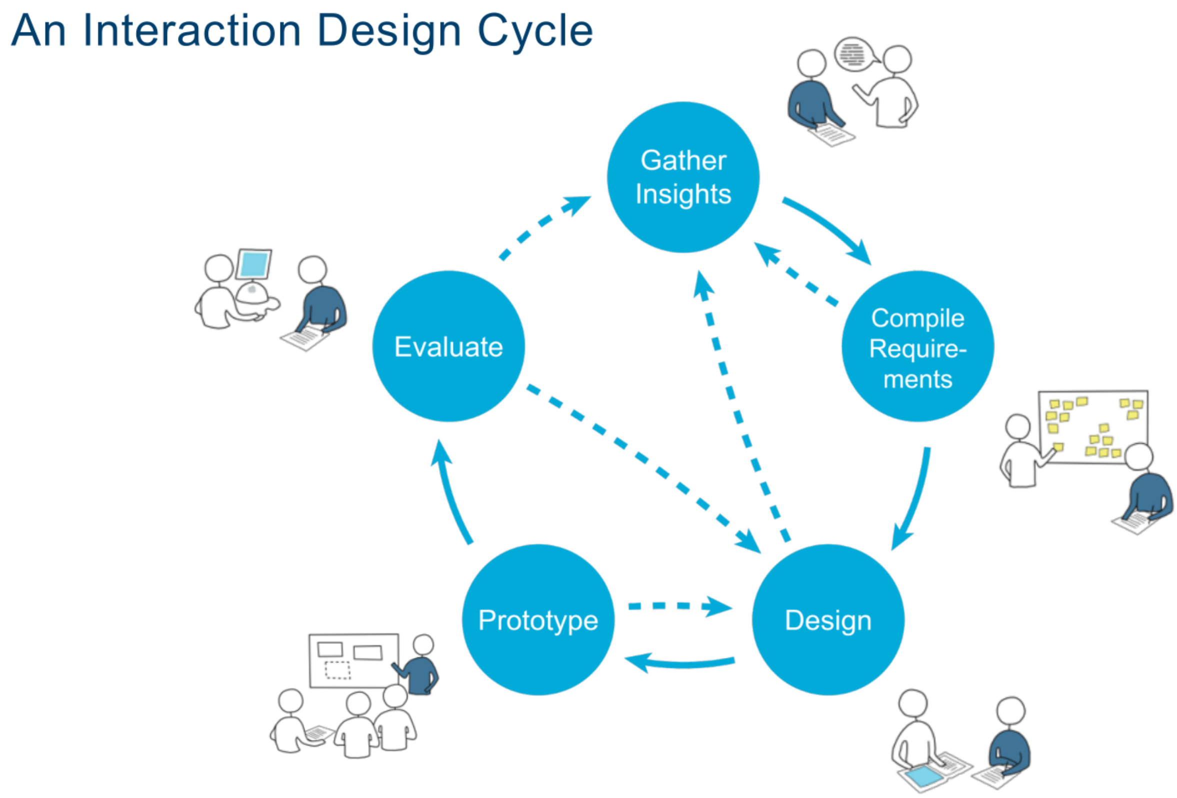


Figure 1: Interaction Design Cycle as presented in the lecture.

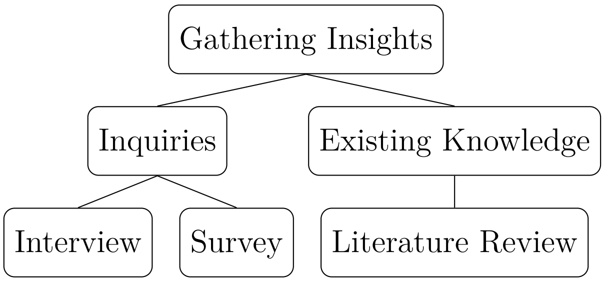
3. **Design:** Create a prototype mockup that fulfills the compiled requirements.
4. **Prototype:** Build a prototype that implements the design.
5. **Evaluate:** Gather feedback from real users interacting with the prototype.

2 Gathering Insights

2.1 Methodology

We designed an interview sheet, later adjusted to double as a survey sheet (Figure 4). The sheet was designed for quick interviews at bus stops. It flows from an "about you" section to public transport use and main difficulties, then asks about apps used and desired improvements. Free-text fields reduce interviewer effect. The third section (main difficulties) was the core.

We also extended our earlier literature research on known public transport issues. The methods used for gathering insights:



We collected 54 responses, 38 from newcomers. Sampling bias was a problem: distributing the survey among private contacts whose newcomer status was known in advance skewed results toward younger age groups and people from Germany. Free-text analysis added effort – some answers were in German, some were hard to interpret.

2.2 Interpretation

2.2.1 Evaluation of Free-Text Responses

Free-text responses were evaluated qualitatively. Responses describing concrete needs, suggestions, or recurring problems became direct input for prototype development. Remarks about missing information, confusing ticket options, or accessibility issues were translated into design requirements. Responses not introducing a new aspect were assigned to existing problem categories to keep the analysis consistent with structured survey data.

2.2.2 Visualizing Results

We built automated scripts to analyze survey responses, cross-referencing residence duration against system understanding, and age against usage frequency.

We generated graphs and charts from the data. Figure 2 shows the hurdles newcomers encounter most often.

3 Compiling Requirements

3.1 Scope Definition

This project scope is limited. The objective is not a production-ready mobility product. The project focuses on human-computer interaction aspects of public transport – how newcomers understand routes, tickets, and transport-related information. Given the timeframe, the prototype prioritizes interaction design, usability, and information communication over complete technical coverage. Several features are simplified or conceptual: the prototype's purpose is to evaluate whether the proposed interface reduces confusion and supports better transport decisions.

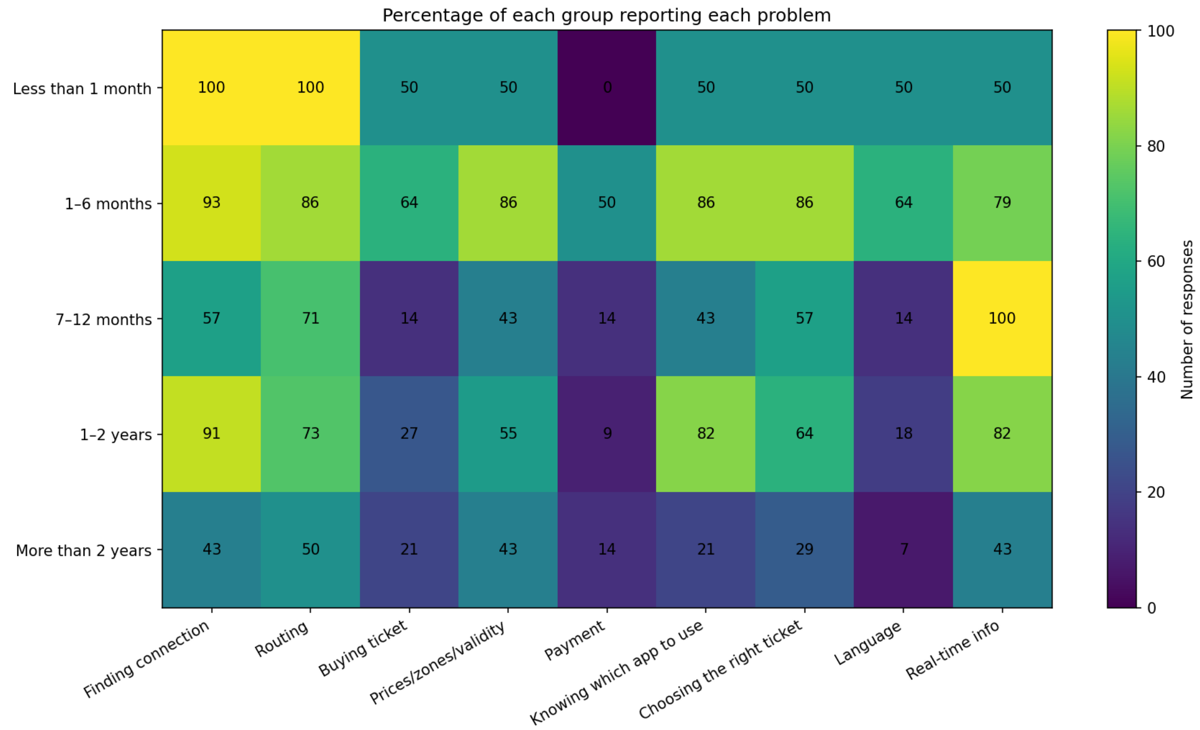


Figure 2: Matrix heatmap showing the percentage of respondents per occurring public transport issue by length of stay in Oldenburg.

The prototype supports public transport use within Oldenburg only. Rides outside Oldenburg are not shown. The available bus and train data is restricted by the VBN network.

The prototype covers route guidance and basic ticket-related support, not a complete transport booking or pricing system. Train pricing uses dummy data; exact train prices are not central to the evaluated concept. Different ticket types are not handled separately. The system indicates whether bus rides are free based on available ticket information.

Some user preferences appear in the interface but are not connected to routing logic. Bicycle and dog options are displayed in the user profile and route preferences but do not change route suggestions. This keeps the technical implementation manageable.

The prototype does not implement an independent routing engine. Route suggestions use available API data and simplified assumptions. Taxi and car routing are not implemented. The prototype is a focused interaction prototype, not a replacement for existing mobility platforms.

3.2 Requirements

The requirements below derive from the major issues we gathered, scoped to the prototype boundaries.

3.2.1 Functional Requirements

ID	Requirement
FR1	The system shall provide public transport route planning for Oldenburg, Germany.
FR2	The system shall allow users to search routes between a start location and a destination and display route details such as transport type (Taxi, Bus, Train), stops, transfers, duration, departure time, and arrival time.
FR3	The system shall integrate the HAFAS Public Transport endpoint to retrieve public transport connection and departure data where available.
FR4	The system shall provide an onboarding process for creating a user profile, including available tickets, preferred transport modes, disabilities or accessibility needs, walking distance preferences, and regular travel habits.
FR5	The system shall personalize route and ticket information based on the user profile, including a compact ticket overview with validity information and basic route suitability guidance.

Table 1: Functional Requirements

3.2.2 Non-Functional Requirements

ID	Requirement
NFR1	The system shall be simple, beginner-friendly, and easy to understand for users with little public transport experience.
NFR2	The system shall present ticket, route, stop, and departure information in a compact and understandable way.
NFR3	The system shall be mobile-friendly, responsive, and usable as an installable Progressive Web App.
NFR4	The system shall keep route, ticket, and departure information as up-to-date as possible within the limitations of the available HAFAS data source.
NFR5	The system shall provide trustworthy and accessible guidance by using clear labels, readable text, sufficient contrast, fallback messages for missing data, and transparent ticket or route information.

Table 2: Non-Functional Requirements

4 Designing

4.1 Design Evaluation and Requirement Mapping

A low-fidelity Figma prototype was created to visualize the user interface before implementation.

Mapping the mockups against the Functional Requirements (FR) and Non-Functional Requirements (NFR) shows which goals were addressed during design and which are delegated to backend implementation.

4.1.1 Functional Requirements Check

The current Figma prototypes focus on the ticket optimization flow, not on complete routing mechanics.

Addressed in Design

- **FR1 (Route Planning for Oldenburg):** The start screen establishes the scope of Oldenburg, for example by presenting journeys such as *Hauptbahnhof to Stadtmuseum* within *Zone Oldenburg I*.
- **FR5 (Personalized Ticket Overview):** An automated verification step ("Automatic check running . . . ") combined with a step-by-step wizard collects traveler information: age, bicycles, dogs, and travel intent (single trip versus commuting). Based on this input, the system generates a compact ticket recommendation on the final screen.

Missing or Delegated to Implementation

- **FR2 (Route Details):** The screens do not display a detailed route breakdown – intermediate stops, transfers, or transport modes. This will be designed separately or integrated alongside the ticket view.
- **FR4 (Onboarding/Profile):** The UI references an "Existing pass check" but the onboarding or profile creation flow is not shown in the current mockups.
- **FR3 (HAFAS API Integration):** As the prototype is static, live API integration cannot be demonstrated visually.

4.1.2 Non-Functional Requirements Check

The design emphasizes usability, accessibility, and clarity.

Addressed in Design

- **NFR1 (Simple and Beginner-Friendly):** The wizard breaks down tariff systems into a sequence of questions: "Who is travelling?" and "What should we optimise?". The reassurance text ("We'll only ask what changes the price") reduces cognitive load.
- **NFR2 (Compact Information):** The final ticket overview highlights price, validity, and zone information using visual cards. The optimized recommendation (e.g., a 4x Ticket) is compared against a standard Single Ticket.
- **NFR3 (Mobile-Friendly):** The prototype uses a mobile viewport with large touch targets, bottom-anchored primary action buttons ("Continue", "Buy ticket"), and vertical scrolling.
- **NFR5 (Trustworthy and Transparent):** Ticket recommendations are explained, price comparisons are shown, and standard high-contrast color schemes (blue and white) are used.

Delegated to Implementation

- **NFR4 (Up-to-date Data):** Departure times and ticket data freshness depends on backend systems and external data sources. This cannot be validated within a static Figma prototype.

4.1.3 Limitations of the Mockup

Static Figma prototypes simulate certain system behaviors. Live data retrieval from the HAFAS system (FR3, NFR4) and real-time profile-based logic cannot be executed. The "Automatic check running" indicator is an optimistic UI placeholder. Dynamic routing and optimization features will be validated during implementation.

5 Prototyping

5.1 Technologies

After gathering insights, compiling requirements, and creating the visual design, we implemented an interactive prototype. The goal was to make the central concept testable: users search a route, understand the connection, and receive ticket or fare guidance.

The prototype is a React and Vite web application. Tailwind CSS handles styling. Leaflet provides map display. The VBN/HAFAS endpoint supplies public transport data. Profile and session data are stored locally in IndexedDB. Application state is managed in a React context storing the selected route, trip data, profile settings, passengers, search results, and ticket-flow answers.

This stack provided the features needed to fulfill the requirements while enabling a PWA.

The code is separated into screen components and domain modules. Screen components render the interface and handle user interaction. Domain modules contain the application logic. This separation keeps visual code independent from transport-specific calculations. Changes to the interface do not affect the core logic.

Passenger validation, fare handling, taxi fare calculation, ticket optimization, geometry handling, and HAFAS product classification are implemented outside the visual components. This makes the logic testable independently from the UI. Taxi fares, fare calculation, passenger validation, geometry, and HAFAS mode parsing are covered by automated tests.

5.2 Implemented Features

Users can search for a connection, inspect route details, and get a ticket recommendation or taxi fare estimate. The route result list shows departure and arrival times, duration, transport lines, transfers, and walking information. The journey detail screen includes a map, timeline, boarding and exit information, and a direction indicator.

The user profile is a central feature. During onboarding, users define whether they travel as an adult or child, which tickets they own, which transport modes they prefer, and whether accessibility support is relevant. The profile can be enabled or disabled in the profile menu. When enabled, it filters route results and influences ticket recommendations.

The profile includes “prefer low-floor vehicles” and “notify driver” options, derived from survey responses about accessibility and insecurity at stops. These are not available in DB Navigator. In the current prototype, they exist in the interface but do not trigger real transport-network functionality.

The prototype includes taxi support. When taxis are enabled in the profile, a taxi option appears at the top of the result list. Selecting it skips the public transport ticket questions and directly shows a fare estimate based on the local Oldenburg taxi tariff, with separate day and night/Sunday rates.

5.3 Prototype Workflow

Workflow starts with onboarding. After profile creation, the user enters a start and destination, selects departure time, and searches for connections. Public transport connections lead to the journey detail screen and then to the ticket flow. The ticket flow asks only trip-relevant questions and shows a recommendation. Taxi connections use a shorter path to the fare estimate.

5.4 Implementation Decisions

Mode preferences are managed in the profile rather than adding visible filters to the route planner. This keeps the search screen less crowded. Details are separated from result cards: cards show key information, the journey detail screen contains the full explanation.

For the map, the prototype uses available journey geometry where possible. If detailed geometry is missing, it falls back to simplified or estimated geometry.

5.5 Requirements Checking

The prototype fulfills the main functional requirements within the defined scope: route planning for Oldenburg and surrounding VBN connections, journey details, HAFAS integration, persistent user profile, and personalized route and ticket information. It addresses the non-functional requirements with compact, mobile-friendly, beginner-oriented presentation.

The requirements are fulfilled at prototype level. The route planner works with available API data but does not replace a full routing engine. Ticket logic is a simplified model sufficient for testing the concept. Profile-based personalization is implemented for visible filtering and recommendation logic, not all operational effects.

6 Technical Limitations

The prototype has several technical limitations from available data sources, project scope, and focus on interaction design. Ticket purchasing is not supported. Implementing it would require collaboration with official ticket providers, including payment systems, fare validation, and issuing infrastructure. The prototype provides ticket guidance instead.

Ticket prices are static and not updated automatically. The prototype is not connected to an official fare database, so prices may become outdated and are illustrative, not legally binding.

Accessibility features are partly conceptual. Notifying bus drivers about wheelchair users is included as an idea without technical functionality. “Prefer low-floor vehicles” and “notify driver” options do not affect routing logic, since comparable functionality is not provided by DB Navigator or the available transport data.

Routing is partly estimated. The prototype relies on API data instead of an independent routing engine. Route suggestions depend on the quality and completeness of the external data source.

7 Evaluation

7.1 Evaluation Method

We conducted a user study using the System Usability Scale (SUS). SUS measures effectiveness, efficiency, and satisfaction through a 10-item Likert scale and yields a single score from 0 to 100.

We chose SUS over the User Experience Questionnaire (UEQ) because UEQ assesses broader dimensions – attractiveness, stimulation, novelty – across 26 items, which exceeds our goals. Our prototype resolves functional barriers for newcomers rather than evoking specific emotional responses. The focused 10-item SUS offered a more direct measurement of core usability.

We created an anonymous Google Forms survey titled "Oldenburg Public Transport App - Usability Study" (the full questionnaire is in Figure 5). The survey had four sections:

- **About you:** Demographic data (age, time in Oldenburg) and transport usage habits (frequency, purpose, apps).
- **Try the app:** Participants opened the prototype and completed three scenarios.
  - **Task A:** Plan a one-way weekday trip from Oldenburg Hauptbahnhof (Hbf) to Lappan (city centre).
  - **Task B:** Read the app's ticket suggestion for the previous connection and pick the ticket they would buy.
  - **Task C:** Adjust their profile to simulate using a four-ride ticket (Vierer-Ticket) and verify the ticket counter decreased.

  After each task, users rated whether they completed it and how easy it was.
- **System Usability Scale (SUS):** Ten standard SUS statements rated on a scale from 1 to 5.
- **Final Thoughts:** Open-ended questions about what worked well and what needed improvement, and whether they would use the app for public transport in Oldenburg.

7.2 Discussion of Results

The 49 participants who tested the application produced a mean SUS score of 70.9 and a median of 72.5. The industry average for SUS is 68. A score of 70.9 places the

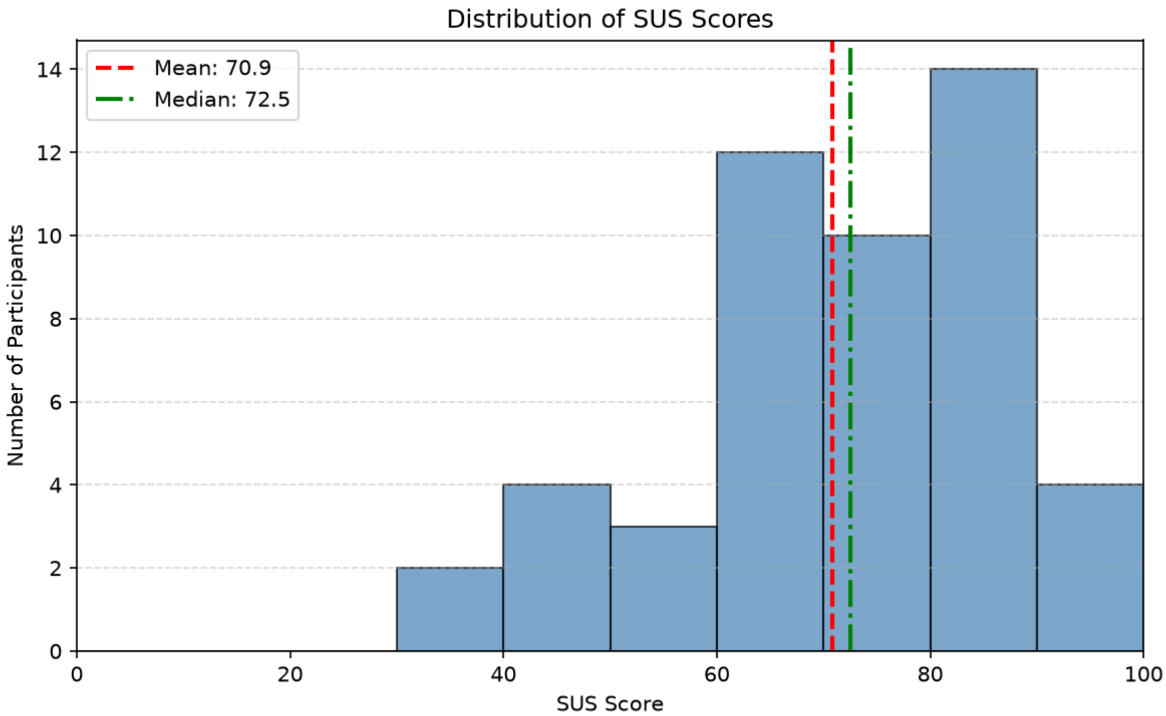


Figure 3: SUS Histogram showing the final score distribution over all 49 users that tested the deployed application.

usability above average, in the "Good" range. The median of 72.5 shows more than half of participants rated the system highly.

Most users scored the application between 60 and 100. The most frequent scores fell in the 80 to 90 range (14 participants). This concentration suggests that the wizard-based ticket selection and compact route overviews reduced cognitive load and addressed the initial pain points identified during insight gathering.

A small cluster of lower scores (30 to 50) indicates room for refinement. These likely correspond to edge cases or technical limitations (static ticket pricing, limited routing data) rather than a failure of the core concept.

7.3 Conclusion

This project applied the Interaction Design Cycle to the challenges newcomers face with public transport in Oldenburg. Surveys and inquiries identified key pain points: ticket complexity and routing confusion. The prototype emphasized clarity, accessibility, and guided ticket selection over an broad feature set. The SUS score of 70.9 confirms the design improved usability. The prototype is conceptual and has technical limitations, but the interface shows that a user-centric design can lower the entry barrier for public transport systems.

7.4 Future Work

Future work includes integrating a live ticket purchasing system through collaboration with transport providers, replacing static pricing and route estimations with a real-time routing engine, implementing accessibility features such as the driver notification system, and expanding the application scope to regional travel outside Oldenburg.

A Survey Data Structure

Sample structure of the JSON data collected during the field surveys.

Listing 1: Sample Survey Data Structure


```
{
    "respondent_id": 1,
    "user_type": "newcomer",
    "difficulties": ["routing", "buying tickets"],
    "usage_frequency": "daily"
}
```

Public transportation, apps, and tickets for newcomers in Oldenburg

This survey is fully anonymous.

1. About you

Age group

Under 18 | 18-25 | 26-35
36-50 | 51-65 | Over 65

Where are you from?

Germany | Another country | Prefer not to say

If another country: country/region (optional)

How long have you lived in Oldenburg?

Less than 1 month | 1-6 months | 7-12 months | 1-2 years
More than 2 years

2. Public transport use

How often do you use buses/trains?

Daily | Several times/week | Rarely | Never

What do you mainly use it for?

Study/school | Work | Shopping/appointments | Leisure/visits
Attending events

At the beginning, how easy was it to understand public transport in Oldenburg?

Very easy | Easy | Okay | Difficult | Very difficult

3. Main difficulties

What is/was difficult?

Finding the right connection
Routing
Buying a ticket
Prices/zones/validity
Payment
Knowing which app to use
Choosing the right ticket
Language
Real-time info

Other difficulty (optional)

4. Tickets and apps

Which apps/websites have you used?

Google Maps | DB Navigator | VBN/local app
Transport website | None | Other

Have you ever been unsure whether your ticket was valid?

Yes | No | Not sure

5. One quick story and improvement

One situation that was confusing or stressful:

What would have helped most?

Would a simple beginner guide help?

Yes | No | Maybe

Figure 4: Survey and interview sheet used for inquiries.

Oldenburg Public Transport App - Usability Study

Thank you for helping us evaluate our prototype of a public transport app for Oldenburg.

What you need: About 8-10 minutes, the prototype open in a browser tab, and a simple journey idea (e.g. "from Oldenburg Hbf to Lappan").

Privacy: This form is anonymous. We do not ask for your personal data.

Tasks: In Section 2 you will be asked to perform three short scenarios on the prototype. The form stays open in a separate tab - switch back when you are done. If you get stuck, do your best and tell us about it in the open feedback box at the end.

* Gibt eine erforderliche Frage an

About you

Demographics and usage habits.

1. 1.1 What is your age group? *

Markieren Sie nur ein Oval.

- Under 18
- 18-25
- 26-35
- 36-50
- 51-65
- Over 65
- Prefer not to say

Figure 5: Complete Google Forms questionnaire used for the usability study and gathering user feedback.

2. 1.2 How long have you lived in Oldenburg / the region? *

Markieren Sie nur ein Oval.

- Less than 1 month
- 1-6 months
- 7-12 months
- 1-2 years
- More than 2 years

3. 1.3 How often do you use public transport in Oldenburg? *

Markieren Sie nur ein Oval.

- Never
- Rarely (a few times a year)
- Several times a month
- Several times a week
- Daily

4. 1.4 For which purposes do you typically use public transport? *

Wählen Sie alle zutreffenden Antworten aus.

- [ ] Work / commute
- [ ] Study
- [ ] Shopping / appointments
- [ ] Leisure / visits
- [ ] Events
- [ ] Sonstiges: ____________

5. 1.5 Which apps or websites do you currently use to plan journeys here?

Wählen Sie alle zutreffenden Antworten aus.

- [ ] Google Maps
- [ ] DB Navigator
- [ ] VBN app / Fahrplaner
- [ ] Local transit app
- [ ] The VBN transport website
- [ ] I do not use any
- [ ] Sonstiges: ____________________

Try the app

Open the prototype in a new browser tab: https://o3l.de/hci/. Keep this form open in a second tab - you will return to it after each task.

Please complete the three tasks in the order shown. The tasks mimic real situations newcomers to Oldenburg often face.

Task A - Plan a one-way trip

Imagine it is tomorrow (weekday) at 14:00 and you want to travel from Oldenburg Hauptbahnhof (Hbf) to Lappan (city centre). Find a valid connection.

6. 2.1 I was able to complete Task A. *

Markieren Sie nur ein Oval.

- ( ) Yes, fully
- ( ) Yes, with some guessing
- ( ) Partially
- ( ) No, I could not

7. 2.2 Rate how easy it was to set the start, destination, date and time. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very easy

Task B - Pick a ticket
Using the same connection as Task A, read which ticket the app suggests and why. If it offers a "DayTicket" or "SingleTicket", pick the one you would actually buy.

8. 2.3 I was able to complete Task B. *

Markieren Sie nur ein Oval.

- ◯ Yes, fully
- ◯ Yes, with some guessing
- ◯ Partially
- ◯ No, I could not

9. 2.4 Rate how easy it was to understand the ticket recommendation. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very easy

Task C - Profile and four-ride ticket (Vierer-Ticket)
Set "Adult" sections to 3 in your Profile. Trigger a free journey and click "Ride without ticket". Check that the counter went from 3 to 2.

10. 2.5 I was able to complete Task C. *

Markieren Sie nur ein Oval.

- Yes, fully
- Yes, with some guessing
- Partially
- No, I could not

11. 2.6 Rate how easy it was to find and use the four-ride ticket counter. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Very						Very easy

12. 2.7 If you answered "Partially" or "No" to any of the tasks above, what blocked you? (Otherwise, leave blank)

System Usability Scale (SUS)

For each of the following statements, mark one circle that best describes your reaction to the prototype you just used.

13. 3.1 I think that I would like to use this app frequently. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro						Strongly agree

14. 3.2 I found the app unnecessarily complex. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro						Strongly agree

15. 3.3 I thought the app was easy to use. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro						Strongly agree

16. 3.4 I think that I would need the support of a technical person to be able to use this app. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro						Strongly agree

17. 3.5 I found the various functions in the app were well integrated. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

18. 3.6 I thought there was too much inconsistency in the app. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

19. 3.7 I would imagine that most people would learn to use this app very quickly. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

20. 3.8 I found the app very cumbersome to use. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

21. 3.9 I felt very confident using the app. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

22. 3.10 I needed to learn a lot of things before I could get going with this app. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly agree

Final Thoughts

23. 4.1 What worked well, and what was confusing or needed improvement?

24. 4.2 Would you use this app in real life for your public transport trips in Oldenburg? *

Markieren Sie nur ein Oval.

- Definitely yes
- Probably yes
- Not sure
- Probably not
- Definitely not

25. 4.3 (Optional) Age / contact for follow-up - leave blank to stay anonymous

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